National University of Computer and Emerging Sciences
Lahore Campus

Database Systems (CS2005) Sessional-1 Exam

Date: Mon, 23 Feb 2026

Course Instructor(s)

IR, ZA, MM, MN, AA

20+3 = 23

Total Time (Hrs.): 1
Total Marks: 25
Total Questions: 4

Roll No Section Student Signature

Instructions: Attempt all the questions in the space provided. Extra sheets will not be checked.

Consider the following basic schema for an App Store system. The APP table maintains information about different applications available on the platform, while the APP_DEVELOPER table maintains the list of all developers.

The ACCOUNT table stores basic information for both users and developers. Each developer is associated with an account in the ACCOUNT table. The attributes DeveloperAccountID and AppID in the APP_DEVELOPER table are the foreign keys that reference in the ACCOUNT and APP table respectively. The ACTIVITY relation records the interaction between accounts and applications, including the download_date and the rating assigned to an app by different accounts. The rating is represented as a numerical value ranging from 1 to 5. Note Download_Date is NOT NULL attribute.

APP

AppID	Name	Price
101	ChatMaster	25000
102	PhotoEdit Pro	50000
103	GameZone	85000
104	FitTracker'	70000
105	LearnSQL	90000
106	MusicStream	100000
107	TravelBuddy	65000

APP_Developers

AppID	DeveloperAccountID
101	2
102	3
101	5
104	7
104	2
105	3
106	5
105	7

Account

AccountID	Name	Country
1	Ali	Pakistan
2	Ahmad	Pakistan
3	Sara	USA
4	John	UK
5	Bilal	UAE
6	Ayesha	Canada
7	Noor	Australia

Activity

ActivityID	AccountID	AppID	Download_Date	Rating
1	2	104	2025-01-10	4
2	4	105	2026-01-10	1
3	6	103	2025-02-10	2
4	2	104	2025-01-10	3
5	4	105	2026-02-10	1
6	3	105	2025-01-10	5

CLO # 4: *Author queries using relational algebra and SQL.*

Q. No 1: Specify the following queries in *SQL*? **(10 points)**

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a) Find the AppID of the apps that are created by two or more developers and are never downloaded.

```
SELECT A.AppID FROM (App A LEFT JOIN Activity AC ON
A.AppID = AC.AppID) INNER JOIN App_Developers AD ON
A.AppID = AD.AppID
WHERE AC.ActivityID IS NULL
AND A.AppID IN (SELECT DISTINCT AppID FROM
App_Developers WHERE DeveloperAccountID IN (
SELECT DeveloperAccountID FROM App_Developer
GROUP BY DeveloperAccountID
HAVING COUNT(*) >= 2));
```

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b) List the names of the Pakistani App Developers who have given a rating of 5 to their own Apps.

```
SELECT A.Name FROM
(Account A INNER JOIN App_Developers AD
ON A.AccountID = AD.DeveloperAccountID)
INNER JOIN Activity AC ON
AD.AppID = AC.AppID AND AD.dev-Acc(1) = AC.Acc(1)
WHERE A.Country = 'Pakistan'
AND AC.Rating = 5;
```

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CLO # 4: *Author queries using relational algebra and SQL.*

Q. No 2: Consider the given database and apply the following operations. State if the operation will be carried out successfully or not. In case of successful operation, indicate the changes that will be made to the above database. Also, state all the integrity constraints violated by each operation, if any. Please note that all operations are independent.

Consider all foreign keys have on delete Cascade and on update Cascade. **(3 points)**

a. Insert into APP_Developers values(108,7)

Referential Integrity Constraint failed.
AppId 108 doesn't exist in parent table.

3

b. Delete from APP where name =' FitTracker'

1 Row with Appname = FitTracker will be deleted from APP.
2 Rows with AppId=104 will be deleted from APP Developers & Activity Table each.

c. Update Account Set AccountID = 8 Where AccountID = 3

Account ID will change from 3 to 8 in Account (1 row).
Similarly, 2 rows in APP_Developer & 1 row in Activity table will be updated.

CLO # 4: *Author queries using relational algebra and SQL.*

Q. No 3: Consider the given above basic schema for an App Store system. **(3 points)**

Add referential integrity constraints: "On Delete Cascade" and "On update NoAction" to the foreign keys to all tables.

3

```
ALTER TABLE APP_Developers
ADD CONSTRAINT FK_AppID
FOREIGN KEY (AppID)
REFERENCES App(AppID)
ON DELETE CASCADE
ON UPDATE NO ACTION;
```

```
ALTER TABLE Activity
ADD CONSTRAINT FK_AppID2
FOREIGN KEY (AppID)
RERERENCES App(AppID)
ON DELETE CASCADE
ON UPDATE NO ACTION;
```

```
ALTER TABLE APP_Developers
ADD CONSTRAINT FK_DevID
FOREIGN KEY (DeveloperAccountID)
RERERENCES Account(AccountID)
ON DELETE CASCADE
ON UPDATE NoACTION;
```

```
ALTER TABLE ACTIVITY
ADD CONSTRAINT FK_DevID2
FOREIGN KEY (Account ID)
REFERENCES Account (AccountID)
ON DELETE CASCADE
ON UPDATE NoACTION;
```

(Assuming that it will work even if we have FK constraint before)
(to edit only ON DELETE/UPDATE settings, we need constraint name)

0006

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CLO # 4: *Author queries using relational algebra and SQL*

Q. No 4: Consider the given database and give the output in the form of DB relation. Also, explain in one sentence what the query is doing. **(9 points)**

```
A) select AP.AppID,Ap.Name from APP AP inner join Activity AC
   on AP.AppID=AC.AppID
   group by AP.AppID,Ap.Name
   having count(Distinct AC.AccountID)>1 and AVG(AC.Rating)>=2
```

AppID	Name
105	LearnSQL

The Query returns ID & name of apps to which more than 1 unique users have intelacted and it has an average rating of 2 or above.

```
B) select Ac.AccountID from Activity AC
   group by Ac.AccountID
   having AVG(AC.Rating)>2
   intersect
   (select A.AccountID from Account A
   where A.Country='Pakistan' or A.Country='USA'
   union
   select AD.DeveloperAccountID from APP_Developers AD
   group by  AD.DeveloperAccountID
   having count(*)>1)
```

Account ID
2
3

The query returns Account ID of users who are either from Pakistan or USA or have worked as a developer in more than 1 apps and they have given an average rating of more than 2 to the apps they have interacted with.

```
C) select A.AccountID,A.Name from (Account A
   left join Activity AC on A.AccountID=AC.AccountID)
   inner join APP_Developers AD on AD.DeveloperAccountID=A.AccountID
   where AC.AccountID is null
   group by A.AccountID,A.Name
   having count(*)>1
```

AccountID	Name
5	Bilal
7	Noor

The query returns AccountID & Name of developers who have developed more than 1 apps and have never interacted with any app. (Only developed no activity/download)